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InterCellar: an R/Shiny app enabling interactive analysis of cell-cell interactions from single-cell RNA sequencing data

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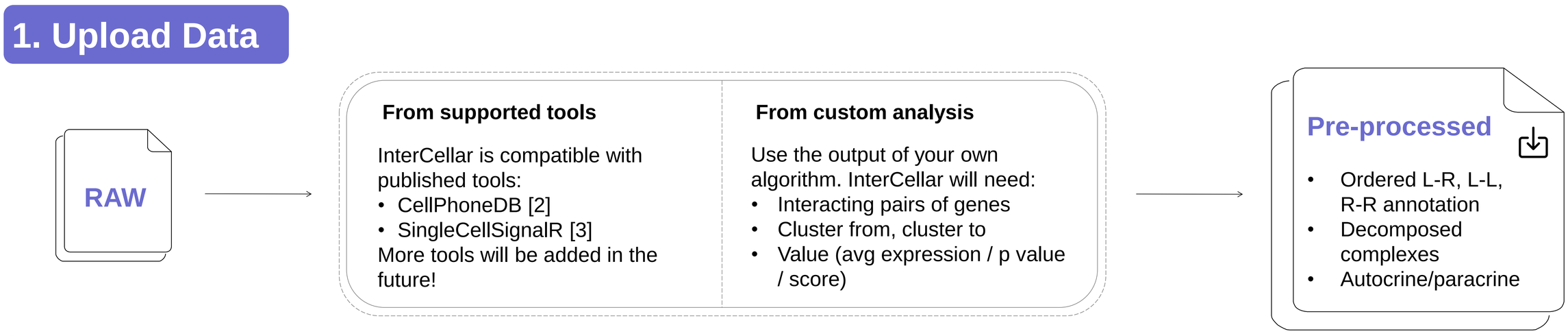
Introduction

One important application of single-cell RNA sequencing (scRNA-seq) is the analysis of cellular interactions. Cell-to-cell communication, mediated by secreted ligands that bind to membrane receptors, is implicated in a variety of physiological and pathological functions. In the last few years, extensive efforts have been made to create comprehensive databases of ligand-receptor interactions, coupled with statistical frameworks able to predict the intercellular communication across multiple cell types. However, to this date only few tools offer functionalities to conduct a downstream analysis of the predicted interactions. Moreover, the complexity of the results obtained, worsened by the high number of possible combinations between cell types present in the data, makes the interpretation lengthy and often biased by previous knowledge of the analyst. Here we present InterCellar, an R/Shiny app that enables the interactive exploration of ligand-receptor interactions in scRNA-seq data. With no programming skills required, researchers are able to perform a downstream analysis which is data-driven and reproducible.

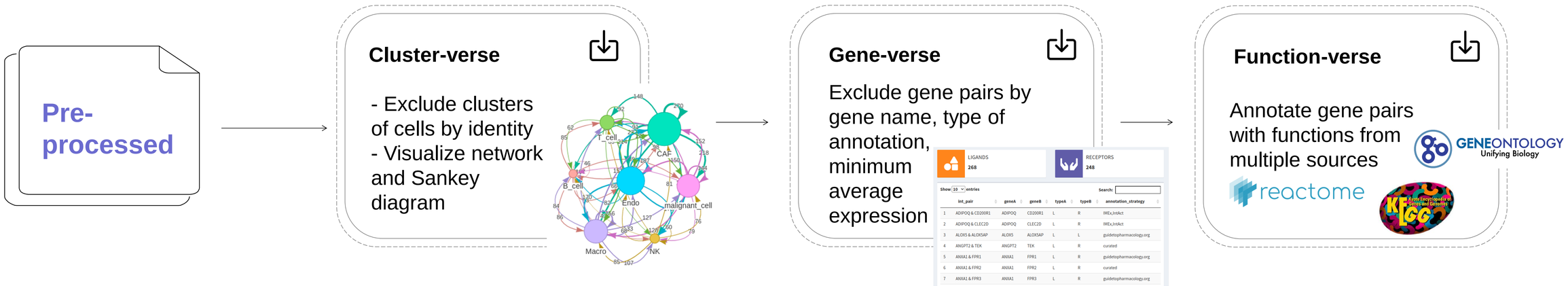
Workflow and features

In the following, an exemplary workflow is presented, based on a single-cell melanoma dataset from [1].

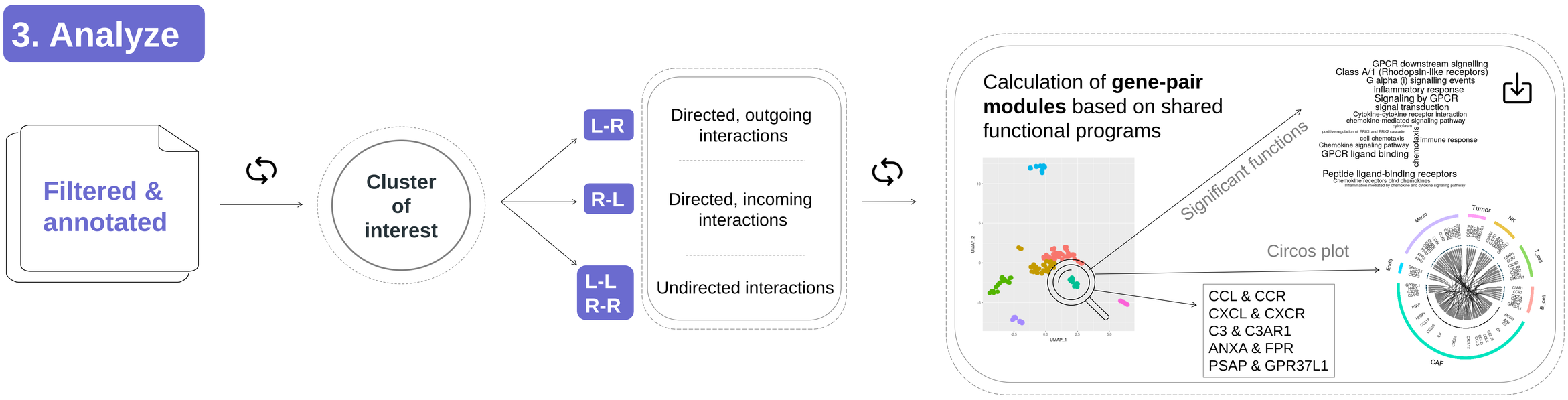
1. Upload Data



2. Universes: filter, annotate and visualize



3. Analyze



Availability

At the time of writing, InterCellar (version alpha) is under active development and only available to selected users.
Stay tuned for the beta release at https://github.com/martaint/InterCellar!

References

[1] Tirosh, Itay, et al. "Dissecting the multicellular ecosystem of metastatic melanoma by single-cell RNA-seq." *Science* 352.6282 (2016): 189-196.
[2] Efremova, Mirjana, et al. "CellPhoneDB: inferring cell–cell communication from combined expression of multi-subunit ligand–receptor complexes." *Nature protocols* 15.4 (2020): 1484-1506.
[3] Cabello-Aguilar, Simon, et al. "SingleCellSignalR: inference of intercellular networks from single-cell transcriptomics." *Nucleic acids research* 48.10 (2020): e55-e55.

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